

Scalability & Future Plan

Date	04 July 2026
Team ID	
Project Name	Smart Lender – AI
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S. No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses a limited dataset for training	Reduces prediction accuracy and generalization for unseen applicant data	High
2	Supports only single-user access at a time.	Cannot efficiently handle multiple concurrent users	Medium
3	Uses local storage and lacks cloud deployment	Limits accessibility, scalability, and data availability	Medium

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited users On a local machine	Deploy the application on cloud platforms such as AWS, Azure, or Render with load balancing to support multiple concurrent users..
2	Data Storage	Uses local CSV	Migrate to scalable databases such as MySQL, PostgreSQL, or MongoDB for efficient storage and retrieval.

3	Performance	Prediction on a local system	Optimize the machine learning model and deploy using cloud infrastructure with REST APIs for faster response times.
4	Security	Basic application security	Implement user authentication, HTTPS, role-based access control, and encrypted data storage.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve the Machine Learning model using larger datasets, advanced feature engineering, and hyperparameter tuning.	1-2 months	Increases approval prediction accuracy, reduces bias, and improves reliability.
Phase 2	Deploy the application on cloud platforms (AWS, Azure, or Render) with a scalable database backend and REST API services.	2-4 months	Enables multi-user access, high availability, better scalability, and centralized data management.
Phase 3	Integrate AI-powered recommendations, real-time credit score updates, interactive dashboards, and explainable AI (XAI) for prediction transparency.	4-6 months	. Provides personalized insights, improves user trust, and enhances the overall user experience.